

Statistics, B.S.

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Statistics, B.S. Program (120 units)

Program Description

Statistics is the science of learning from data. Majors study collecting and analyzing all kinds of data, and reporting the results of the analysis. Designs of Experiments are used to plan data collection in experimental settings, such as in medical research. Surveys are used to collect opinions from samples in populations, such as in political polling. Observational data is collected commonly by business as part of standard business processes, such as in website analytics. Statistical methods are used to process data into easily read visual displays, and into summary statistics for interpretation, decision making, and reporting.

Statistics has been a rapidly growing science for many years and has many uses in other fields, such as Econometrics, Business Analytics, Machine Learning, and Visualization. Statistics is widely used by businesses and government organizations to understand changes in the economy and to make forecasts about future events based on past patterns in their data. Statistics is used to model relationships between variables and to predict future values of output variables from input variables. Statistical methods draw accurate inferences about large groups on the basis of a representative sample from the group or population. Descriptive Statistics are used to summarize and visualize the collected data.

The methodology of statistics can be adapted to many types of problems. Due to the extensive development of computers and the collection of large databases, the need for statistical techniques has greatly expanded in recent years. A society like ours, which has become increasingly dependent on its data, has a growing need for statisticians.

Student Learning Outcomes

Students graduating with a B.S. in Statistics from Cal State East Bay will be able to:

1. Apply basic computational skills in descriptive statistics and data visualization, hypothesis testing, confidence intervals, modeling and error analysis, including the use of large data sets.
2. Analyze data using appropriate software, including cloud-based software, and to interpret results covering descriptive statistics and data visualization, hypothesis testing, confidence intervals, modeling and error analysis, including the use of

large data sets.

3. Communicate to others results involving descriptive statistics and data visualization, hypothesis testing, confidence intervals, modeling and error analysis using reproducible research best practices.
4. Acquire data using methods of design of experiments, survey sampling, or observational data, including data scraping and data wrangling from open source data and free data sources.

Program Roadmaps

These program roadmaps represent recommended pathways through the program. Please see an advisor to create an education plan that is customized to meet your needs.

4-Year Freshmen Roadmaps

- [Statistics, B.S. \(Non Concentration Students\)](#)
- [Data Science Concentration, Statistics, B.S.](#)

2-Year ADT Roadmaps

- [Statistics, B.S. ADT \(Non Concentration Students\)](#)

Career Opportunities

Actuary • Biostatistician • Business Analyst • Census Analyst • Credit Analyst • Customer Analytics Specialist • Data Analyst • Data Manager • Data Scientist • Data Visualization Designer • Educational Researcher • Financial Modeler • Information Systems Analyst • Marketing Analytics Specialist • Quality Control Specialist • R programmer • Research Statistician • Sales Analyst • Statistical Programmer • Sports Analyst • Statistician • Survey Designer • Systems Analyst • Teacher

Preparation

- For Advanced Placement course equivalencies, see the [Undergrad Orientation & Enrollment](#) chapter.
- The theory of statistics rests on a mathematical foundation. Even for statisticians whose primary interest is in applications, challenging mathematical problems arise continually. Calculus and computer programming are required for the major. An elementary course in linear algebra is highly recommended.
- Courses in areas to which statistics is applied will also be helpful. Examples are business, biology, chemistry, geography, social sciences, psychology, and physics.
- Community college students are advised to complete the lower division requirements for the major before transferring to Cal State East Bay

Credit/No Credit Courses

It is the policy of the department that no course taken on a “CR/NC” basis may be applied toward any of the Upper Division Core requirements. Individual petitions for waiver of this policy will be treated according to the same procedures as petitions to waive other degree requirements.

Double Major

A knowledge of statistical methods is of increasing importance to students in many areas, especially the physical, computing, decision, environmental, biological, and social sciences and administrative studies (including business, health care, and education). A second major in statistics along with a major in one of these areas may result in better preparation for employment or graduate study. The following three factors combine to make such a double major feasible in many cases:

1. The Statistics major allows for a relatively large number of completely unrestricted elective units. The student may elect courses required for the other major among these.

2. Up to fifteen units in certain areas outside of Statistics may be applied toward the Statistics degree. (See Area of Emphasis requirement, 4.)

3. Some other majors require Statistics courses that also count towards the Statistics major (e.g., [STAT 303](#), [STAT 320](#), [STAT 330](#), etc.).

Contact an advisor to develop a clear plan for completing the major.

Degree Requirements Unit-Outline

- A baccalaureate of science degree requires a total of 120 units:
 - The major requirements consist of 49-50 units;
 - General Education (GE) & Graduation Requirements (GR) consist of 52 units;
 - Free Electives may consist of 18-19 units (actual number of free elective units may depend on GE/GR units).

Note: *It may be possible to double-count units within the graduation requirements or a course may satisfy both a graduation requirement and a major requirement. Students should contact their program advisors for information.*

Statistics Major Requirements (49-50 units)

Please consult an advisor in the department for clarification and interpretation of the major requirements.

Introductory Core

Total 10-11 units

- [MATH 130 - Calculus I](#) Units: 4 ; Breadth Area: GE-2

Select two (2) courses from the following courses for 6-7 units:

- [CS 101 - Computer Science I](#) Units: 4
- [MATH 131 - Calculus II](#) Units: 3
- [MATH 215 - Introduction to Linear Algebra](#) Units: 3
- [STAT 206 - Public Health Informatics and Technology](#) Units: 3
- [STAT 215 - Introduction to Data Science](#) Units: 3

Advanced Core

The following courses for 24 units are required as outlined below:

Take all of the following:

- [STAT 330 - Statistical Inference](#) Units: 3
- [STAT 331 - Introduction to Analysis of Variance](#) Units: 3
- [STAT 432 - Introduction to Linear Regression and Logistic Regression](#) Units: 3

Select one (1) of the following:

- [STAT 320 - Introduction to Probability Theory I](#) Units: 3
- [STAT 321 - Probability Through Simulation](#) Units: 3

Select four (4) Elective Courses from the following:

- [STAT 306 - Public Health Informatics: The Role of Data](#) Units: 3
- [STAT 312 - Data Analysis with Python](#) Units: 3

- [STAT 340 - Data Science Ethics](#) Units: 3
- [STAT 351 - Sampling Procedures for Surveys](#) Units: 3
- [STAT 406 - Emerging Technologies for Public Health](#) Units: 3
- [STAT 416 - Public Health Maps and Spatial Analysis for Health Equity](#) Units: 3
- [STAT 450 - Introduction to R for Data Science](#) Units: 3
- [STAT 451 - Introduction to Data Visualization](#) Units: 3
- [STAT 452 - Introduction to Statistical Learning](#) Units: 3
- [STAT 473 - Introduction to Nonparametric Statistics](#) Units: 3
- [STAT 475 - Introduction to Stochastic Processes](#) Units: 3
- [STAT 481 - Bayesian Statistics](#) Units: 3

Emphasis Coursework

Only students NOT completing the Data Science concentration are required to complete an emphasis area. Complete one (1) of the following for 15 units:

- Fifteen (15) units of approved Statistics courses in addition to those used for the requirements above. [MATH 230](#) may be included in these 15 units and is especially recommended for students wishing to apply to the master's degree program in Statistics.
- **OR**, Fifteen (15) units of approved Mathematics courses in addition to those used for the requirements above. MATH 230 must be included in these 15 units.
- **OR**, Fifteen (15) units of approved Computer Science courses in addition to those used for the requirements above.
- **OR**, Fifteen (15) units of approved courses in an approved area. Areas currently approved include the following: Anthropology, Biological Sciences, Business Administration, Chemistry, Economics, Geography, Geological Sciences, Health Sciences, Physics, Psychology, Sociology. For other areas, contact the Department of Statistics and Biostatistics. To gain departmental approval, these courses must include at least one upper division course and be judged to constitute a coherent program of study. (With the approval of the department, upper division Statistics courses not counted above, except STAT 310 and STAT 303, may be applied toward these fifteen units.)

Other Undergraduate Degree Requirements

In addition to major requirements, every student must also complete the University's baccalaureate requirements for graduation, which are described in the [Undergrad Baccalaureate & Program Requirements](#) chapter of this catalog.

Department Courses Listed by Course Type

Statistics, Biostatistics, and Data Science Courses

Department of Statistics and Data Science

[Go to information for this department.](#)

Courses

Biostatistics: Graduate

- [BSTA 661 - Categorical Data Analysis](#) Units: 2
- [BSTA 662 - Survival Analysis](#) Units: 2
- [BSTA 663 - Clinical Trials in the Pharmaceutical and Biomedical Industries](#) Units: 2
- [BSTA 665 - Survival Data Analysis using SAS](#) Units: 4

- [BSTA 668 - Longitudinal Data Analysis](#) Units: 2
- [BSTA 697 - Issues in Biostatistics](#) Units: 1-3

Data Science

- [DATA 105 - Math for Data Science 1](#) Units: 5 ; Breadth Area: GE-2
- [DATA 130 - Data Science Profession](#) Units: 2
- [DATA 205 - Math for Data Science 2](#) Units: 3
- [DATA 312 - Data Analysis with Python](#) Units: 3
- [DATA 340 - Data Science Ethics](#) Units: 3
- [DATA 492 - Capstone Project](#) Units: 3

Statistics

- [STAT 100 - Elements of Statistics and Probability](#) Units: 3 ; Breadth Area: GE-2
- [STAT 100A - Statistics Principles](#) Units: 1
- [STAT 100E - Elements of Statistics and Probability with Extra Support](#) Units: 4 ; Breadth Area: GE-2
- [STAT 101 - Statistics in Everyday Life](#) Units: 3 ; Breadth Area: GE-2
- [STAT 101A - Everyday Statistics Principles](#) Units: 1
- [STAT 110 - Elements of Statistics for Business and Economics](#) Units: 3
- [STAT 125 - Introduction to Statistics in Social Sciences](#) Units: 3 ; Breadth Area: GE-2
- [STAT 125A - Statistics Principles for Social Sciences](#) Units: 1
- [STAT 130 - Data Science Profession](#) Units: 2
- [STAT 206 - Public Health Informatics and Technology](#) Units: 3
- [STAT 215 - Introduction to Data Science](#) Units: 3
- [STAT 303 - Statistical Methods in Biology](#) Units: 3
- [STAT 303A - Biostatistics for Health Science](#) Units: 3
- [STAT 306 - Public Health Informatics: The Role of Data](#) Units: 3
- [STAT 310 - Statistical Methods in the Social Sciences](#) Units: 4
- [STAT 312 - Data Analysis with Python](#) Units: 3
- [STAT 316 - Statistics and Probability for Science and Engineering](#) Units: 3
- [STAT 320 - Introduction to Probability Theory I](#) Units: 3
- [STAT 321 - Probability Through Simulation](#) Units: 3
- [STAT 330 - Statistical Inference](#) Units: 3
- [STAT 331 - Introduction to Analysis of Variance](#) Units: 3
- [STAT 340 - Data Science Ethics](#) Units: 3
- [STAT 351 - Sampling Procedures for Surveys](#) Units: 3
- [STAT 398 - Internship](#) Units: 1-3
- [STAT 406 - Emerging Technologies for Public Health](#) Units: 3
- [STAT 416 - Public Health Maps and Spatial Analysis for Health Equity](#) Units: 3
- [STAT 432 - Introduction to Linear Regression and Logistic Regression](#) Units: 3
- [STAT 450 - Introduction to R for Data Science](#) Units: 3
- [STAT 451 - Introduction to Data Visualization](#) Units: 3
- [STAT 452 - Introduction to Statistical Learning](#) Units: 3
- [STAT 473 - Introduction to Nonparametric Statistics](#) Units: 3
- [STAT 474 - Introduction to Time Series and Forecasting](#) Units: 3
- [STAT 475 - Introduction to Stochastic Processes](#) Units: 3
- [STAT 481 - Bayesian Statistics](#) Units: 3

- [STAT 490 - Independent Study](#) Units: 1-4
- [STAT 492 - Capstone Project](#) Units: 3
- [STAT 495 - Data Analysis with SAS](#) Units: 3
- [STAT 497 - Issues in Statistics](#) Units: 3
- [STAT 498 - Internship](#) Units: 1-3

Statistics: Graduate

- [STAT 605 - Teaching Mathematics at the University Level](#) Units: 1
- [STAT 620 - Probability and Statistical Theory](#) Units: 4
- [STAT 630 - Statistical Methods](#) Units: 4
- [STAT 633A - Generalized Linear Models I](#) Units: 4
- [STAT 633B - Generalized Linear Models II](#) Units: 4
- [STAT 640 - Advanced Statistical Theory](#) Units: 4
- [STAT 640A - Advanced Statistical Theory I](#) Units: 2
- [STAT 640B - Advanced Statistical Theory II](#) Units: 2
- [STAT 641 - Bootstrapping Methods](#) Units: 2
- [STAT 650A - Advanced R for Data Science](#) Units: 2
- [STAT 650B - Python for Data Science](#) Units: 2
- [STAT 651 - Data Visualization](#) Units: 2
- [STAT 652 - Statistical Learning](#) Units: 2
- [STAT 653 - Statistical Natural Language Processing](#) Units: 2
- [STAT 654 - Introduction to Applied Deep Learning](#) Units: 2
- [STAT 660 - Advanced SAS I](#) Units: 2
- [STAT 661 - Categorical Data Analysis](#) Units: 2
- [STAT 671 - Advanced JMP](#) Units: 2
- [STAT 673 - Nonparametric Statistical Methods](#) Units: 2
- [STAT 674 - Time Series](#) Units: 2
- [STAT 675 - Advanced Stochastic Processes and Simulation](#) Units: 2
- [STAT 676 - Advanced Probability](#) Units: 4
- [STAT 690 - Independent Study](#) Units: 1-4
- [STAT 692 - Comprehensive Exam Review](#) Units: 2
- [STAT 694 - Applied Research in Statistics and Biostatistics](#) Units: 1-4
- [STAT 697 - Issues in Statistics](#) Units: 2-4
- [STAT 698 - Internship](#) Units: 1-3